Detector Design Spec 2 — 4H-SiC Microdosimeter (sensitive-volume geometry)

For: foundry hand-off (FBK Schottky / ST JBS / other) From: 4H-SiC detector TCAD feasibility study Date: 2026-06-16 Status: GEOMETRY-ONLY physics specification (NOT a mask set; junction type NOT modeled)

> Scope and honesty boundary. This document specifies the sensitive-volume geometry and electrode layout of a 4H-SiC microdosimeter, sized from TCAD depletion and CCE-vs-bias (laterally uniform model). It does NOT specify the junction type or its electrical behavior. Our simulator has no Schottky and no JBS model — so the barrier height, built-in potential, leakage, and C-V of a metal/JBS front contact are explicitly left to the foundry (FBK Schottky, ST JBS). The geometry is portable across junction types; the depletion-bias numbers below assume a p-n-like junction electrostatics and must be re-derived by the fab for a Schottky/JBS contact (see §4).

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1. Device concept

Array of small sensitive volumes (SVs) in a 4H-SiC epitaxial layer, each read out through a front electrode. The microdosimetric geometry (SV dimensions, depth, electrode layout) is what we specify. The junction that forms the collecting contact — Schottky (FBK) or junction-barrier-Schottky (ST) — is the foundry's choice and is dropped into this geometry.

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2. Sensitive-volume geometry spec (baseline, p-n electrostatics assumed)

Parameter	Recommended value	Basis / note
SV depth (epi thickness)	10 µm (baseline)	default stack (profile fitted to the digitised measured C-V)
SV half-width	50 µm (→ 100 µm SV)	baseline choice only; NOT discriminated by the model (no edge physics)
Graded epi donor profile N_D(y)	N_D_junction 8.19e15, N_D_bulk 3.38e13, L 0.46 µm; graded n^-/n^+ transition over the last 1.14 µm to 4.5e16	C-V fit (DEFAULT_STACK; N_D_bulk below the documented 0.5–1e14; transition not from SIMS)
Device stack	p^+ 0.3 µm / n^- epi / n^+ substrate	owner-confirmed; cathode on the n^+
Operating reverse bias	≥ −10 V (practical; gate ≥ 3.3 V)	1.2 × V_fd = 3.28 V; reach-through 3.10 V; −10 V keeps margin for the fit's doping uncertainty (DESIGN-1 §2)
Full-depletion voltage V_fd (10 µm)	≈ 2.7 V	analytic graded profile to the transition start (assumes p-n electrostatics)
Validity gate	|V_bias| ≥ 1.2 × V_fd and CCE ≥ 0.95	Mj-3 validity gate; 20 % margin = ±20 % doping tolerance (DESIGN-1)

Alternative SV depths (model V_fd, p-n electrostatics):

Epi thickness	V_fd	Recommended operating bias	Note
5 µm	~0.7 V	≥ −10 V	lowest-bias option; epi holds ~42 % of the alpha's pairs
10 µm	~2.7 V	≥ −10 V (gate 3.3 V)	baseline (C-V-fitted profile)
20 µm	~11.5 V	≥ −14 V (1.2 × V_fd)	converges (no wall with the n^+ substrate)

(5 and 20 µm carry the 10 µm device's fitted profile and 1.14 µm transition: an extrapolation. With the previous abrupt-n^+ default the three V_fd were 4.0 / 9.9 / 33 V.)

No solver wall on the default stack. The cathode sits on the n^+ substrate and devsim converges to ≥ −100 V in 2D. The ~−50 V limit of earlier versions belonged to the legacy model (ohmic cathode directly on the n^- epi, `etna.legacy_config()` / `microdosimetric_sweep(nplus_thickness_cm=None)`): there the contact-cell field grows past reach-through until devsim stops converging (≈ 4.7 × 10^4 V/cm) and such rows are NaN with `failure_reason` "outside model validity". Never punch-through, never breakdown.

Validity gate (audit Mj-3): only configurations that are fully depleted with margin ($|V_{bias}| \geq 1.2 \times V_{fd}$, covering the ±20 % doping tolerance) and above the CCE floor (0.95 at both center and edge) are quoted. Partially depleted points are excluded because their CCE is < 0.95.

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3. What the geometry deliverable gives the fab

- SV depth, half-width, and the laterally-uniform graded doping target at which the SV is fully depleted at the operating bias.
- Full-depletion voltage and recommended operating bias for a p-n-type junction.
- CCE → 1 (> 0.96) when fully depleted (DD model; consistent with the 1D twin). CCE is relative to the e-h pairs generated inside the simulated epi, not to the full alpha deposit: the model alpha (5.486 MeV, NIST ASTAR Bragg curve in SiC, 18.4 µm range, since 2026-10-20) leaves ~43 % of its pairs in a 10 µm epi at normal incidence (5 µm: ~20 %, 20 µm: ~100 %; the earlier 15 µm flat box gave ~83 % / ~42 % / ~100 %); collected/deposited ≈ that fraction × CCE. The lateral scans report both (`in_mesh_generation_fraction`, `q_collected_over_deposited`).

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4. Junction type — NOT modeled (fab fills in), and why the bias numbers shift

The collecting junction (FBK Schottky / ST JBS) is not in our simulator. Specifically NOT provided and to be supplied/measured by the fab:

- Schottky/JBS barrier height φ_B, built-in potential, image-force lowering.
- Leakage of the metal/JBS junction (Schottky leakage ≫ p-n; thermionic + TFE).
- C-V and the bias needed to fully deplete the SV.
- JBS p^+ grid geometry and edge termination.

Critical: our V_{fd} and W(0 V) numbers assume p-n junction electrostatics. A Schottky/JBS front contact has a different built-in potential, so the bias required to fully deplete the same geometry will differ. The fab must re-derive the depletion bias from its measured barrier height and C-V. The SV geometry is portable; the bias numbers in §2 are not — treat them as the p-n reference, not a spec for the Schottky/JBS device.

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5. Caveats — do NOT read as committed specs

- No Schottky/JBS electrical parameters — barrier height, leakage, C-V, turn-on are not modeled; do not expect them here.
- Dark current / leakage: not specified. Our p-n leakage is a calibration input (N_t fitted so the model equals the upper bound "< 18 pA up to −60 V" at −60 V, 4 mm², 300 K; heuristic bias shape; effective "TAT" generation with the Hurkx (1 + Γ) field enhancement with an assumed tunnelling mass; assumed $n_i(T)$ temperature scaling) and does not apply to a Schottky/JBS junction. The fab must measure leakage for its junction type.
- No rated breakdown / max voltage. No breakdown model. (The ~−50 V convergence limit of the legacy model, cathode on the n^- epi, was a solver artefact, not a rating.)
- Tissue-equivalence (κ) and microdosimetric spectra are NOT in this spec. The tissue-equivalence correction depends on stopping-power data we have flagged as not foundry-ready (Phase 27 work). This document is geometry only; it makes no claim about lineal-energy spectra.
- SV widths other than 50 µm half-width are not quoted — the model cannot rank width; choosing one needs edge physics (field crowding, termination) or measurement.

- “CCE uniformity” is not a modelled result (corrected 2026-10-16). The 2D model is laterally uniform by construction (doping depends on depth only, both contacts span the full width, both lateral sides are mirror boundaries), so edge CCE equals centre CCE and the “uniformity” gate (edge & centre ≥ 0.95) is satisfied trivially. The edge/centre differences reported before this date (~1e-4) were a numerical artefact of the I_generated quadrature (it scaled with mesh spacing), not edge physics. Real edge behaviour (field crowding, termination, guard rings, surface charge, lateral doping) is not modelled, so no sweep ranking by uniformity says anything about the layout.

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6. What is solid (model geometry targets, within the limits above)

The SV depth, the full-depletion voltage and operating-bias window (for p-n electrostatics, with the 1.2 × V_fd margin), and CCE → 1 when fully depleted (relative to in-epi generation). The 50 µm half-width is a baseline choice that the model does not discriminate (no edge physics). The junction electrical behavior is the foundry’s to define and measure.